

RedrobRank

Agentic Recruiter OS

Grounded candidate discovery for Redrob's Data & AI Challenge

Team: Syntheon

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Problem: Intelligent Candidate Discovery

AI Talent Discovery Is Not Keyword Search

1. Keyword stuffing

AI terms appear without real engineering depth.

2. Ghost candidates

Profiles may be inactive, unavailable, or low-response.

3. Domain mismatch

Support, accounting, HR, content, or operations roles can contain AI language.

4. Hallucinated reasons

A ranking system loses trust when the reason does not match candidate facts.

RedrobRank optimizes relevance, trust, and reproducibility.

From 100K Profiles to a Grounded Top-100

1. Extract Evidence

Title, YOE, location,
career, skills, Redrob
behavioral signals.



2. Score Fit

AI/retrieval, production,
evaluation, systems,
YOE, location,
behavior.



3. Filter Risk

Honeypots, keyword
stuffers, non-fit roles,
manager-only.



4. Explain Output

Every reason
references actual
candidate facts.

100K processed

100 final rows

CPU-only

No LLM/API

Validator passed

What the Senior AI Engineer JD Needs

✓ Must-have signals

- Retrieval / vector search / semantic search
- Ranking / recommendation systems
- LLM / RAG / fine-tuning / NLP
- Production ML / MLOps
- Evaluation: NDCG, MRR, A/B testing
- Python + data/systems infrastructure

■ ■ Risk signals

- Non-technical role mismatch
- Prompt-only or keyword-stuffed profiles
- Manager-only with weak technical depth
- Pure research without production evidence
- Weak availability or response behavior

Deterministic Multi-Signal Ranking

Scoring Stack:

1. TF-IDF lexical recall
2. Structured AI skill score
3. Skill trust: proficiency, duration, endorsements
4. Career relevance with recency weighting
5. Production + evaluation evidence
6. Behavioral availability
7. Risk penalties
8. RRF fusion + candidate_id tie-break

No black-box ranking. No hidden API calls. Same input, same output.

Every Recommendation Is Auditable

Reasoning uses actual:

- Current title
- YOE
- Location
- Matched AI/retrieval terms
- Production/eval evidence
- Redrob behavioral signals

Metrics & Proof:

- Official validator: Passed
- Internal validator: Passed
- Forensic audit: Passed
- Hallucination failures: 0
- Unique scores: 74
- Unique reasonings: 100

No LLM-generated justifications. Explanations are deterministic and evidence-grounded.

End-to-End Workflow

Candidate JSONL → Stream Reader → Feature Extractor → Risk Detector



TF-IDF + Structured Scorer → RRF Ranker → Grounded Reasoning



Top-100 CSV → Validator → Dashboard / Sandbox

Local-First, Privacy-First Architecture

CLI Ranker:

rank.py, stream reader, feature extractor, scorer, CSV writer, reports

Audit Layer:

validator, forensic audit, security scans, quality reports

Interfaces:

React/Vite dashboard, Streamlit sample sandbox

No external APIs

No keys

No LLM calls

No full dataset browser upload

Final Output Surfaces Real AI Profiles

100,000 processed | 100 ranked | 74 unique scores | 100 unique reasonings
0 hallucination failures | CSV validator passed | Under 5 MB output

CAND_0018499	Senior ML Engineer	7.2 YOE	Noida
CAND_0005260	Senior NLP Engineer	5.2 YOE	Chennai
CAND_0046525	Senior ML Engineer	6.1 YOE	Pune
CAND_0081846	Lead AI Engineer	6.7 YOE	Jaipur
CAND_0042506	Search Engineer	4.2 YOE	Mumbai

Evaluator-Proof by Design

Tech:

- Python
- scikit-learn TF-IDF
- JSONL streaming
- React/Vite
- Streamlit

Quality:

- pytest, coverage
- ruff
- forensic audit
- validator checks

Security:

- Bandit, pip-audit, npm audit
- no secrets
- no raw data
- no LLM/API

Built for reproducibility, not demo theater.

Final Submission Package

GitHub Repo: <https://github.com/Sauravssoni/Screener>

Ranked CSV: `submissions/submission.csv`

Deck PDF: `docs/RedrobRank_Official_Submission_Deck.pdf`

Methodology PDF: `docs/RedrobRank_Methodology.pdf`

Audit: `reports/final_ranking_audit.md`

Dashboard: `npm run dev`

Sandbox: `streamlit run sandbox_app.py`

**RedrobRank is a reproducible, auditable TalentOps engine —
not just a ranking script.**